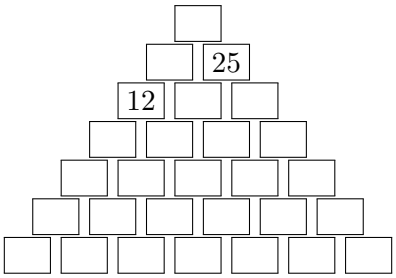
2025 SSMO Tiebreaker Round Solutions

SMO Team

Tiebreaker Round Problem 1: In a triangular grid, each cell must contain exactly one number such that:

- Each cell in the bottom row contains either 0 or 1.
- Each cell not in the bottom row contains the sum of the two numbers in the cells directly below it.

Determine the number in the topmost cell.



Answer: $\boxed{51}$

Solution: Let the numbers in the bottom row be $b_0, b_1, \ldots, b_6$ from left to right. Note that, like in Pascal's triangle, each cell contains a sum of the bottom row numbers weighted by binomial coefficients. So, the two given cells give

$$b_0 + 4b_1 + 6b_2 + 4b_3 + b_4 = 12 \quad \text{and} \quad b_1 + 5b_2 + 10b_3 + 10b_4 + 5b_5 + b_6 = 25.$$

From the second equation, $b_1 + b_6$ is a multiple of 5, so $b_1 = b_6 = 0$. Then, the first equation becomes $b_0 + 6b_2 + 4b_3 + b_4 = 12$, which forces $b_2 = b_3 = 1$ and $b_0 = b_4 = 1$. Substituting into the second equation, we have $5 + 10 + 10 + 5b_5 = 25 \implies b_5 = 0$. So, the number in the topmost cell is

$$b_0 + 6b_1 + 15b_2 + 20b_3 + 15b_4 + 6b_5 + b_6 = 1 + 15 + 20 + 15 = \boxed{51}.$$

Tiebreaker Round Problem 2: Determine the largest positive integer n satisfying

$$\frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+2026} - \sqrt{n+2025}} > 2.$$

Answer: $\boxed{674}$

Solution: Rationalizing the numerator and the denominator, we have

$$\frac{\sqrt{n+1}-\sqrt{n}}{\sqrt{n+2026}-\sqrt{n+2025}} = \frac{\sqrt{n+2026}+\sqrt{n+2025}}{\sqrt{n+1}+\sqrt{n}},$$

so the inequality is equivalent to

$$\sqrt{n+2026}+\sqrt{n+2025} > \sqrt{4n+4}+\sqrt{4n}.$$

If $n \geq 675$, then $n+2026 \leq 4n+4$ and $n+2025 \leq 4n$, so the inequality fails. If $n = 674$, then the inequality becomes $\sqrt{2700}+\sqrt{2699} > \sqrt{2700}+\sqrt{2696}$, which is true. So, the answer is $\boxed{674}$.

Tiebreaker Round Problem 3: Find the number of functions $f : \{1,2,3,4,5,6,7,8\} \to \{1,2,3,4\}$ such that $f(2f(x)) = f(x)$ for all $x \in \{1,2,3,4,5,6,7,8\}$.

Answer: $\boxed{1616}$

Solution: Let S be the set of values taken by f. Since $f(2f(x)) = f(x)$ for all x, we need $f(2v) = v$ for every $v \in S$, and conversely, this is enough for the condition to hold. If $|S| = k$, there are $\binom{4}{k}$ ways to choose S. Then, $f(2v)$ is fixed for each of the k values $v \in S$, and each of the other $8-k$ inputs can be sent to any element of S, giving k^{8-k} functions. So, the answer is

$$\binom{4}{1}\cdot 1^7 + \binom{4}{2}\cdot 2^6 + \binom{4}{3}\cdot 3^5 + \binom{4}{4}\cdot 4^4 = 4+384+972+256 = \boxed{1616}.$$